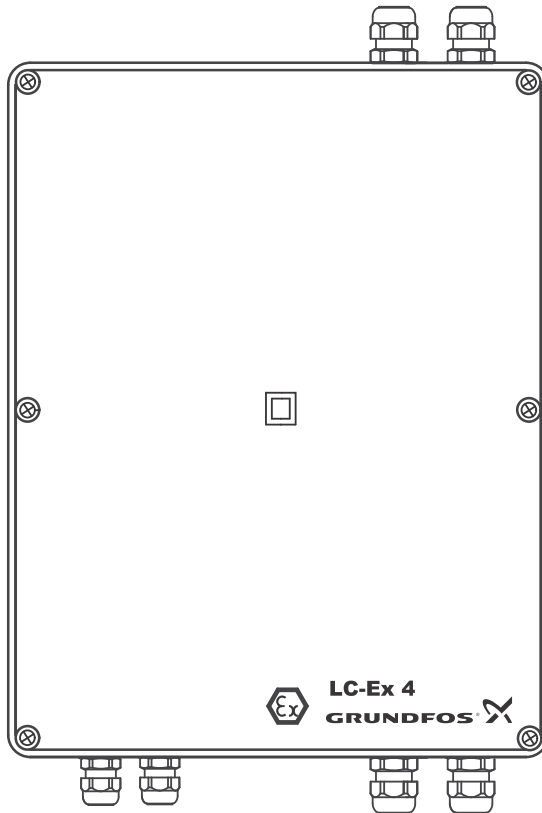


LC-Ex4

Installation and operating instructions



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Declaration of conformity

GB: EC declaration of conformity

We, Grundfos, declare under our sole responsibility that the products LC-Ex4, to which this declaration relates, are in conformity with these Council directives on the approximation of the laws of the EC member states:

- ATEX Directive (94/9/EC).
- Standards used: EN 60079-0:2009 and EN 60079-11:2007.
Applies only to products intended for use in potentially explosive environments, II (1) G, equipped with the separate ATEX approval plate and EC-type examination certificate.

Further information, see below.

Certificate number: DEMKO 11 ATEX 1047766X.

DK: EF-overensstemmelseserklæring

Vi, Grundfos, erklærer under ansvar at produkterne LC-Ex4 som denne erklæring omhandler, er i overensstemmelse med disse af Rådets direktiver om indbyrdes tilnærmelse til EF-medlemsstaternes lovgivning:

- ATEX-direktivet (94/9/EF).
- Anvendte standarder: EN 60079-0:2009 og EN 60079-11:2007.
Gælder kun produkter til eksplosionsfarlige omgivelser, II (1) G, med et separat ATEX-godkendelsesskilt og EF-typeprøvningscertifikat.

Yderligere oplysninger, se nedenfor.

Certifikatnummer: DEMKO 11 ATEX 1047766X.

DE: EG-Konformitätserklärung

Wir, Grundfos, erklären in alleiniger Verantwortung, dass die Produkte LC-Ex4 auf die sich diese Erklärung bezieht, mit den folgenden Richtlinien des Rates zur Angleichung der Rechtsvorschriften der EU-Mitgliedsstaaten übereinstimmen:

- ATEX-Richtlinie (94/9/EG).
- Normen, die verwendet wurden: EN 60079-0:2009 und EN 60079-11:2007.
Gilt nur für Produkte, die für den Gebrauch in potentiell explosiver Umgebung nach II (1) G bestimmt und mit einem separaten ATEX-Typenschild und einem EG-Prüfzeugnis ausgestattet sind.

Weitere Informationen, siehe unten.

Prüfzeugnisnummer: DEMKO 11 ATEX 1047766X.

Bjerringbro, 15th January 2012



Svend Aage Kaae
Technical Director
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Person authorised to compile technical file and
empowered to sign the EC declaration of conformity.

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Notified body No: 0539.

Manufacturer: GRUNDFOS A/S, Poul Due Jensens Vej 7, DK-8850 Bjerringbro, Denmark.

English (GB) Installation and operating instructions

Original installation and operating instructions.

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Warning

Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

1. Symbols used in this document



Warning

If these safety instructions are not observed, it may result in personal injury.



Warning

If these instructions are not observed, it may lead to electric shock with consequent risk of serious personal injury or death.



Warning

These instructions must be observed for explosion-proof pumps. It is advisable also to follow these instructions for standard pumps.



Caution

If these safety instructions are not observed, it may result in malfunction or damage to the equipment.



Note

Notes or instructions that make the job easier and ensure safe operation.

2. General description

The Grundfos LC-Ex4 unit is an intrinsically safe Zener barrier. The unit ensures that signals to and from the float switches never exceed a current or voltage which may cause an explosion.

Warning

The unit has been certified under the DEMKO 11 ATEX 1047766X and complies with the standards EN 60079-0:2009 and EN 60079-11:2007.

Safety class: II (1) G [Ex ia Ga] IIC.



3. Application

The LC-Ex4 unit is used together with the controllers LC 108 and LCD 108 controlling pumps with float switches installed in wastewater environments where explosive gases may be present.

The unit is only suitable for operation with float switches. The basic functions of the LC 108 and LCD 108 remain unchanged.

3.1 Ambient temperature

Ambient temperature during storage and operation:
 $-25^{\circ}\text{C} \leq T_a \leq +50^{\circ}\text{C}$.

4. Electrical connection

The electrical connection must be carried out in accordance with local regulations and special local Ex regulations, if any.



Warning
Install the unit outside the potentially explosive area.



Warning
As the LC-Ex4 and FB-Ex4 have a safety barrier without galvanic separation, an earth connection is necessary for safety reasons.

4.1 Electrical data

Um	250 V	Po	0.4 W
Uo	16.9 V	Io	94 mA
Co	305 nF	Lo	1 mH
Max. Lo/Ro ratio		71 µH/Ohm	

The LC-Ex4 unit has two cables which must be connected to the LC 108 or LCD 108 controller inputs for float switches.

The float switches must be connected to the inputs of the LC-Ex4, see the wiring diagram in fig. B, page 16.

When connecting the float switches, note whether the float switch should close or open the contact to start the pump.

See also section "Electrical connection" in the installation and operating instructions for the LC 108 or LCD 108 controller.

See also the mounting sketch in fig. A, page 15, and the wiring diagram in fig. C, page 17.

Caution *The unit must not be exposed to direct sunlight.*

4.2 Installation

Intrinsically safe wiring must be installed in accordance with the requirements of EN 60079-14 (Explosive atmospheres - Part 14: Electrical installations design, selection and erection).

Place the LC-Ex4 in vertical position. See fig. A, page 15.

Make sure that the cable diameter and cable gland match.

The cable glands of the earth cable: Pg 7, cable diameter of Ø3 to Ø6.5.

The cable glands of the float switches: Pg 9, cable diameter of Ø4 to Ø8.

Note *The cable sheath must be intact all way through the cable gland and minimum 20 mm inside the box.*

Do not strip more conductor insulation and cable sheath than required to fit the terminals inside the box.

5. Maintenance

Under normal operating conditions, the unit is maintenance-free, see section 6. *Fault finding chart.*



Warning
Make sure that no work is carried out in potentially explosive environment.



Warning
Before starting work on the product, switch off the power supply. Make sure that the power supply cannot be accidentally switched on.

5.1 Replacement of parts

The Zener barrier unit or the box including the Zener barrier unit can be replaced.

Note *The Zener barrier unit must not be repaired.*

5.2 Product numbers

Unit	Product number
FB-Ex4, Zener barrier unit	96538241
LC-Ex4, box including the Zener barrier unit	96440300

6. Fault finding chart

Fault	Cause	Remedy
1. Not functioning.	a) In case of too high input voltage, strokes of lightning, faulty installation, etc., the fuse may be blown and/or other components may be damaged.	When the unit is intact, a resistance lower than 500 Ω must be measured between the terminals • 1 and 12 • 2 and 22 • 3 and 32 • 4 and 42. See fig. C, page 17. The resistance must be 0 Ω between the terminals • PE (controller side) and 31 • PE (controller side) and 41. The resistance must be lower than 500 Ω between the terminals • AC and 11. If the measured resistances are not correct, the FB-Ex4 unit is defective and must be replaced, see section 5.2 <i>Product numbers</i>.

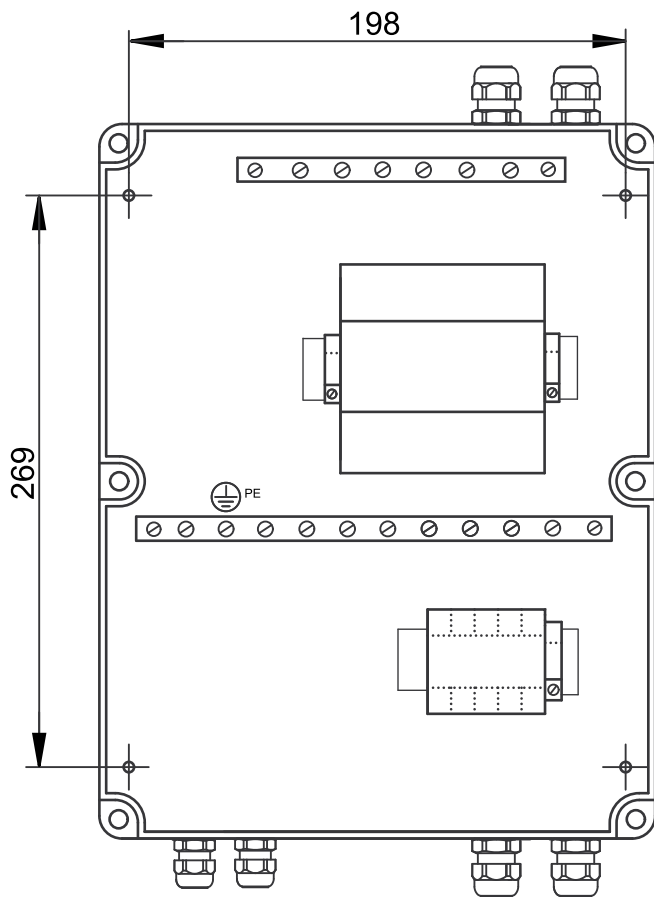
7. Disposal

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.

Subject to alterations.

Fig. A



TM01 9634 2300

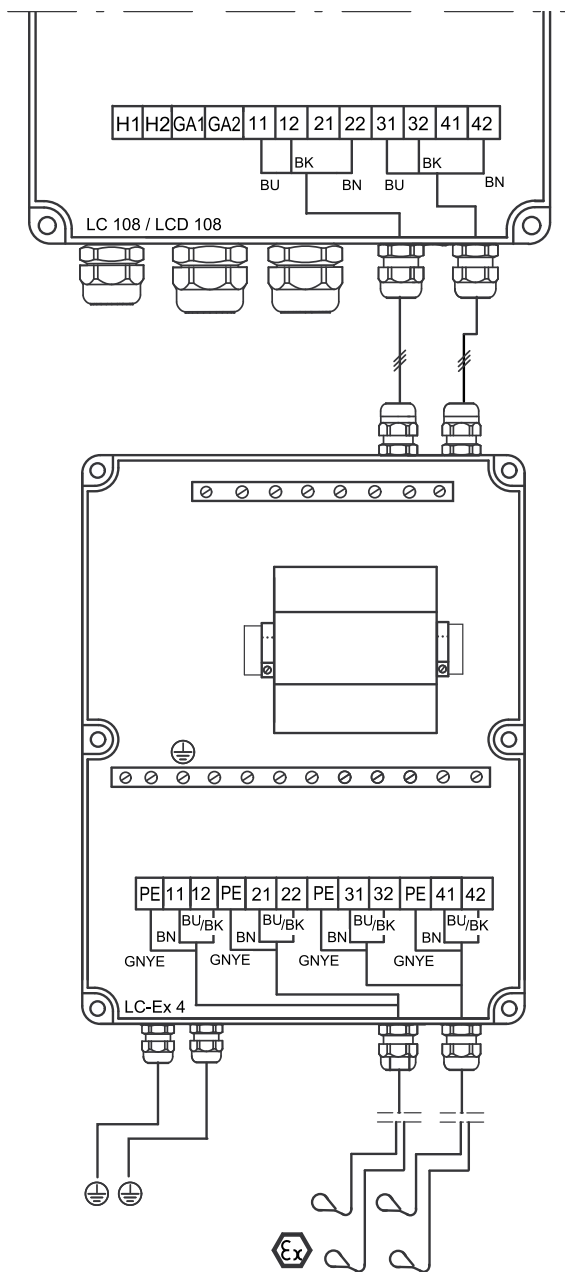
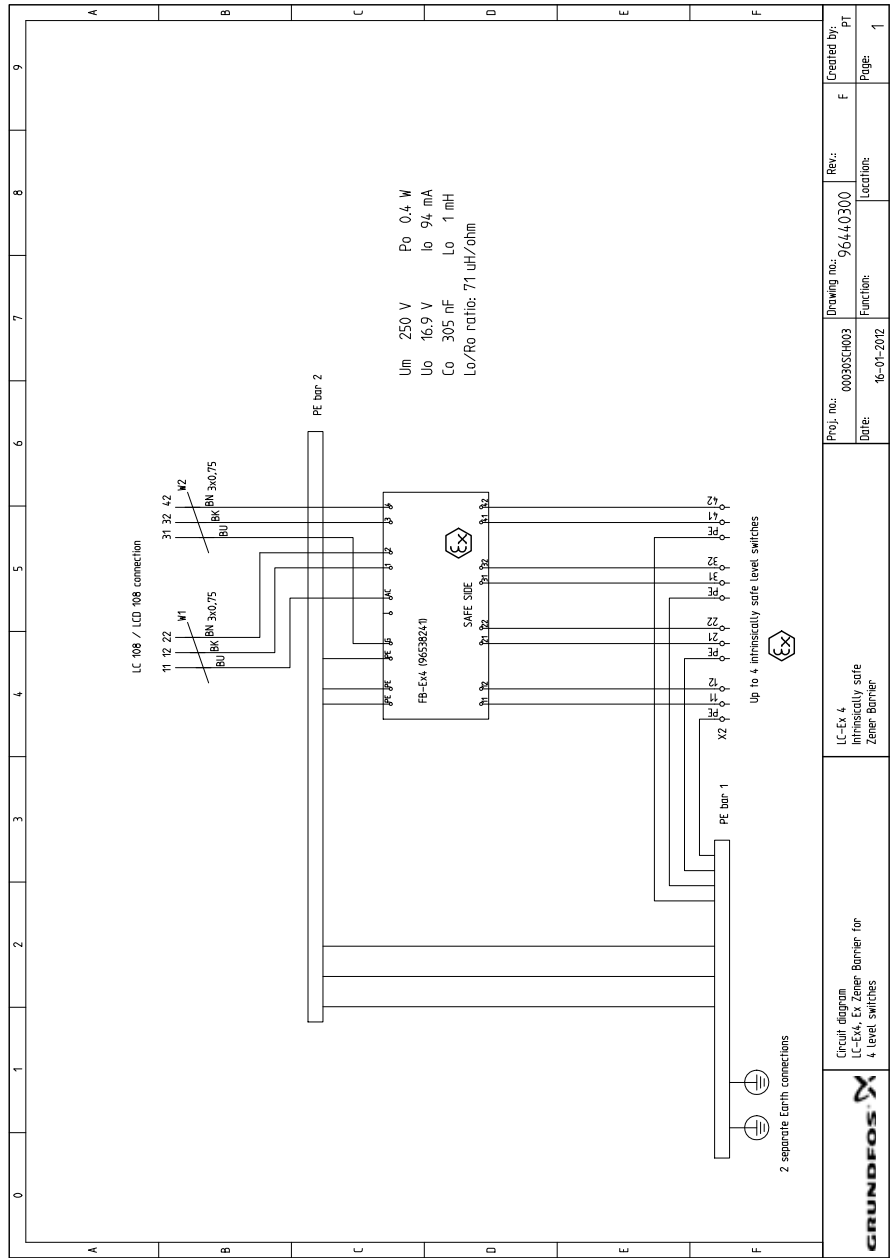


Fig. C



TM02 9200 2104

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